Ranjan Kumar

Final year postgraduate student / MS by Research
Mechanical Engineering, Indian Institute of Technology Kanpur
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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2025 – Present	MS by Research	IIT Kanpur	7.6/10.00
2016 – 2021	B.Tech	GCE, Gaya	7.52/10.00
2013 – 2015	Class XII	GBSS, Khizersarai, Gaya	73.60%
2013	Class X	High School Chandauti, Gaya	76%

Achievements & Extra-Curricular

- Secured a Grade 'A' in Machining Science II.
- Earned a Grade 'A' in the 'AutoCAD - Mechanical & Project' training program at the MSME TRTC, Patna.

Research Projects

Evaluation and optimization of a Percutaneous Mechanical Circulatory Support (interventional blood pump) based on hydraulic & hemocompatibility performance *(Apr '26 – Ongoing)*

Supervisor: Prof. Niraj Sinha, Dept. of Mechanical Engineering, IIT Kanpur

- Perform comprehensive pump characteristics testing to obtain pVAD performance in realtime conditions, perform pressure sensor calibration, & realtime data acquisition system in MATLAB [DAQ Toolbox] and LabVIEW.
- Currently performing CFD analysis in ANSYS for computational H-Q characteristics against in-vitro test rig data through a mesh independence study and SST $k-\omega$ turbulence modelling.

Internship / Experience

Project Technician – *Indian Institute of Technology Guwahati* *(Aug 2024 – Aug 2025)*

- Reconstructed 30+ patient-specific 3D CAD models from clinical CT scans using SimVascular, employing image segmentation, centerline extraction, and 2D lofting techniques, and done CFD simulations to analyze velocity gradients, pressure variations.

Key Projects

Design and Thermal Analysis of Conformal Cooling Channels in injection molding system [ME-670] *ANSYS Fluent*

- Designed & simulated straight vs spiral conformal cooling channels mold in ANSYS 2025; used LMTD method for U evaluation as per Ansys Fluent workflow, demonstrated superior thermal performance of conformal cooling over conventional straight-drilled channels with reduced cycle time and uniform mould temperature, giving a more efficient design tool.

Implicit FDM Solution of Nonlinear Unsteady Diffusion with Radioactive Decay [ME-630]

- Solved 1D transient diffusion-reaction with radioactive decay using fully implicit Finite Difference Method.
- Developed MATLAB code implementing Gauss-Seidel iterative solver for the resulting tridiagonal system; tested grid independence on 201 and 1001 nodes with grid Fourier number = 0.1.

Three-Point Bending Test Simulation using ANSYS Workbench [Self project]

- Performed static structural FEA in ANSYS on a 3-point bending beam using asymmetric frictionless contacts and mesh convergence; validated extracted stress and deformation metrics against Euler-Bernoulli theory.

Relevant Skills

- **Languages/Utilities:** C/C++, Arduino ide, MATLAB, LabView, Autodesk Inventor, AutoCAD, Ansys, Simvascular, Openfoam, ParaView, LaTeX, MS Office, Origin, Ubuntu, Data acquisition and processing (by NI cDAQ).

Relevant Courses

Additive Manufacturing (ME670)	Machining Science 2 (ME662)
Experimental Method in Thermal Sciences (ME649)	Mathematics for Engineers (ME681)
CFD and Heat Transfer (ME630)	

Online Courses & Certifications

- **MATLAB Onramp Academy Certification** – MathWorks *[mathworks.com/verify]*
- **ANSYS Workbench : Learn Structural Analysis** – Udemy *[udemy.com/verify]*
- **AutoCAD Mechanical & Project (Grade A)** – MSME TRTC, Patna *Jul '19*

Positions of Responsibility

Teaching Assistant (ME 212 & ME 261) – Dept. of Mechanical Engineering, IIT Kanpur *(Jan '26 – Present)*

Hobbies

- Photography, Badminton, Swimming, Travelling, Cooking